

24UAM511 - NATURAL LANGUAGE PROCESSING

QUESTION BANK

AI & ML Engineering · III Year · Academic Year 2026

Course: Laboratory Integrated Theory (LIT) PCC · L-T-P-C: 2-0-2-3

Part A — Two-Mark Questions with Answers (20-25 per unit)

Part B — Sixteen-Mark Questions with Answer Hints (6-8 per unit)

Text Books

T1: Raymond S. T. Lee, "Natural Language Processing: A Textbook with Python Implementation", 2nd Edition, Springer Nature, 2025.

T2: Daniel Jurafsky and James H. Martin, "Speech and Language Processing", 3rd Edition, Prentice Hall, 2024.

UNIT I — INTRODUCTION TO NLP

Part A — Two Mark Questions with Answers

Q1. Define Natural Language Processing (NLP).

NLP is a branch of Artificial Intelligence concerned with enabling computers to understand, interpret, generate, and manipulate human (natural) language in both written and spoken forms.

Q2. List any four major challenges in NLP.

Ambiguity (lexical, syntactic, semantic), non-standard language/spelling variations, segmentation of continuous speech/text, idioms and figurative language, coreference resolution, and world-knowledge requirements.

Q3. What is ambiguity in NLP? Give an example.

Ambiguity occurs when a word, phrase, or sentence has more than one possible interpretation. Example: "I saw the man with a telescope" can mean the observer used a telescope, or the man had a telescope.

Q4. Differentiate Natural Language Understanding (NLU) and Natural Language Generation (NLG).

NLU converts natural language input into a structured, machine-interpretable representation (analysis), while NLG converts structured/machine representations into natural language output (synthesis).

Q5. What is a language model (LM)?

A language model assigns a probability to a sequence of words, i.e., it predicts how likely a sentence or the next word in a sequence is, based on the language's structure and usage patterns.

Q6. Differentiate grammar-based and statistical language models.

A grammar-based LM uses explicit hand-written rules (e.g., context-free grammar) to decide well-formedness, while a statistical LM learns word/sequence probabilities from corpus data without explicit rules.

Q7. Define Regular Expression (RE) with an example.

A regular expression is a formal notation for specifying a set (language) of character strings using literals and metacharacters. Example: `[A-Za-z]+` matches one or more alphabetic characters.

Q8. What is a Finite-State Automaton (FSA)?

An FSA is an abstract machine with a finite number of states, a start state, one or more final states, and transitions between states on input symbols, used to recognize regular languages.

Q9. Differentiate a Deterministic FSA (DFA) and Non-deterministic FSA (NFA).

In a DFA, each state has exactly one transition per input symbol, giving a unique path; in an NFA, a state may have zero, one, or multiple transitions for the same symbol, allowing multiple possible paths.

Q10. What is English morphology?

English morphology is the study of the internal structure of words and how they are formed from smaller meaningful units called morphemes (e.g., root, prefix, suffix).

Q11. Define morpheme. Differentiate free and bound morphemes.

A morpheme is the smallest meaning-bearing unit of language. A free morpheme can stand alone as a word (e.g., "book"), while a bound morpheme must attach to another morpheme (e.g., "-ing", "un-").

Q12. Differentiate inflectional and derivational morphology.

Inflectional morphology changes grammatical features (tense, number, case) without changing word class (e.g., cat → cats); derivational morphology creates a new word, often changing its class (e.g., happy → happiness).

Q13. What is a Finite-State Transducer (FST)?

An FST is a finite-state machine with two tapes (input and output) that maps strings of one alphabet to strings of another, used for morphological analysis/generation (e.g., surface form ↔ lexical form).

Q14. How is an FST used for morphological parsing?

An FST combines a lexicon transducer (stem + morphological features) with rule transducers (spelling/orthographic rules) so that an input surface word can be mapped to its stem plus grammatical tags, and vice versa for generation.

Q15. Define tokenization.

Tokenization is the process of splitting a stream of text into meaningful units called tokens — typically words, numbers, or punctuation marks — for further processing.

Q16. Mention two challenges in tokenization.

Handling contractions/clitics (e.g., "don't" → "do" + "n't"), multi-word expressions (e.g., "New York"), hyphenated words, and language-specific issues such as absence of white-space word boundaries (e.g., Chinese).

Q17. Differentiate non-word errors and real-word errors in spelling correction.

Non-word errors produce a string that is not a valid word (e.g., "recieve" for "receive"); real-word errors produce a valid but unintended word (e.g., "there" typed for "their").

Q18. What is minimum edit distance?

Minimum edit distance is the smallest number of insertion, deletion, and substitution operations required to transform one string into another; it is used to measure string similarity, as in spelling correction.

Q19. How is minimum edit distance computed?

It is computed using dynamic programming: a table $D[i][j]$ is built where $D[i][j] = \min(D[i-1][j]+1, D[i][j-1]+1, D[i-1][j-1]+\text{cost})$, with $\text{cost} = 0$ if characters match, else 1 (or 2 for substitution in Levenshtein variants).

Q20. Find the minimum edit distance between "cat" and "cot".

Only one substitution ($a \rightarrow o$) is needed, so the minimum edit distance is 1.

Q21. What is the Noisy Channel Model used for in spelling correction?

It models a misspelled word as the "true" word passed through a noisy channel that introduces errors; correction is done by finding the word w that maximizes $P(w) \times P(\text{observed} | w)$ using Bayes' rule.

Q22. Differentiate stemming and lemmatization.

Stemming crudely chops word endings using heuristic rules to get a root form that may not be a real word (e.g., "studies" $\rightarrow$ "studi"), while lemmatization uses vocabulary and morphological analysis to return the valid dictionary base form (lemma), e.g., "studies" $\rightarrow$ "study".

Q23. Name the four levels of linguistic analysis.

Phonology/Morphology, Syntax, Semantics, and Pragmatics/Discourse.

Q24. Define a corpus in NLP.

A corpus is a large, structured collection of naturally occurring text or speech, often annotated, used to train and evaluate NLP models.

Q25. Differentiate "type" and "token" in text processing.

A token is each individual occurrence of a word/symbol in a text, while a type is a distinct/unique word form; e.g., "the cat sat on the mat" has 6 tokens but 5 types ("the" repeats).

Part B (16 Marks)

Q1. Explain the origins and challenges of Natural Language Processing in detail.

Hint: Trace NLP history (rule-based → statistical → neural eras); discuss challenges — ambiguity (lexical, syntactic, semantic, pragmatic), scale of language data, variability/creativity of language, idioms, world knowledge, low-resource languages. Use examples for each challenge.

Q2. Explain grammar-based and statistical language models with suitable examples.

Hint: Define LM; describe grammar-based LM (CFG-based well-formedness checking) vs statistical LM (n-gram probability estimation, MLE); compare strengths/weaknesses; give a small worked probability example.

Q3. Describe regular expressions and Finite-State Automata. Explain how FSAs are used to recognize regular languages.

Hint: Define RE syntax (literals, disjunction, Kleene star, anchors); define FSA 5-tuple ($Q, \Sigma, \delta, q_0, F$); show a state-transition diagram for a simple pattern (e.g., recognizing "baa+!"); explain DFA vs NFA equivalence.

Q4. Explain English morphology and describe how Finite-State Transducers are used for morphological analysis.

Hint: Define morphemes, inflectional/derivational morphology; explain lexicon FST + rule FST (two-level morphology, cascade); trace parsing of a word like "cats" or "foxes" through the transducer with orthographic rules.

Q5. Explain the process of tokenization and describe the issues involved in tokenizing English text.

Hint: Define tokenization; discuss rule-based vs statistical tokenizers; issues — punctuation, contractions, hyphenation, abbreviations, multi-word expressions, case-folding; briefly mention sentence segmentation.

Q6. Explain the different types of spelling errors and describe algorithms for detecting and correcting them.

Hint: Non-word vs real-word errors; detection via dictionary lookup / n-gram analysis; correction via minimum edit distance candidate generation + Noisy Channel Model ($P(w)$ language model prior $\times P(x|w)$ error model); worked example.

Q7. Explain the minimum edit distance algorithm with a dynamic programming example.

Hint: State recurrence relation, initialize base cases, build the DP table for a pair of strings (e.g., "intention" → "execution"), trace backpointers to show the alignment/edit operations, state final distance value.

Q8. Discuss the linguistic basics relevant to NLP (phonetics/phonology, morphology, syntax, semantics, pragmatics, discourse) with examples.

Hint: Briefly define each level with one example sentence analyzed at each level to show how meaning/structure is built up from sounds to discourse.

UNIT II — WORD LEVEL ANALYSIS

Part A — Two Mark Questions with Answers

Q1. What is an n-gram?

An n-gram is a contiguous sequence of n items (words or characters) from a text; e.g., a bigram uses 2 consecutive words and a trigram uses 3, used to estimate the probability of a word given previous words.

Q2. Define unsmoothed (Maximum Likelihood Estimation) n-gram probability.

The unsmoothed probability of a word given its history is estimated by relative frequency: $P(w_n|w_{n-1}) = \text{Count}(w_{n-1}, w_n) / \text{Count}(w_{n-1})$, computed directly from corpus counts.

Q3. Why is smoothing needed in n-gram language models?

Smoothing is needed because MLE assigns zero probability to unseen n-grams, which is unrealistic; smoothing redistributes some probability mass from seen to unseen events to avoid zero probabilities.

Q4. Explain Laplace (Add-one) smoothing.

Add-one smoothing adds 1 to every n-gram count before normalizing, so $P(w_n|w_{n-1}) = (\text{Count}(w_{n-1}, w_n) + 1) / (\text{Count}(w_{n-1}) + V)$, where V is vocabulary size.

Q5. What is Good-Turing smoothing?

Good-Turing smoothing re-estimates the probability of n-grams with frequency r using the count of n-grams that occur r+1 times, reassigning probability mass from seen to unseen events based on frequency-of-frequency statistics.

Q6. What is perplexity? How is a language model evaluated using it?

Perplexity is the inverse probability of the test set, normalized by the number of words; it is the weighted average branching factor. A lower perplexity indicates a better language model.

Q7. Differentiate intrinsic and extrinsic evaluation of a language model.

Intrinsic evaluation measures the model directly using a metric like perplexity on held-out data; extrinsic evaluation measures the model's effect on the performance of an end task (e.g., speech recognition word error rate).

Q8. What is backoff in n-gram language modeling?

Backoff uses the highest-order n-gram available for a context; if that n-gram was not seen, it "backs off" to a lower-order (n-1)-gram, applying discounting so probabilities still sum to 1.

Q9. What is interpolation in n-gram language modeling? How does it differ from backoff?

Interpolation always combines (weighted-sums) probability estimates from all orders (unigram, bigram, trigram, etc.) simultaneously, whereas backoff uses only the highest-order n-gram with data and falls back only when needed.

Q10. Define word classes (parts of speech).

Word classes, or parts of speech, are categories of words that share similar grammatical properties, such as noun, verb, adjective, adverb, pronoun, preposition, conjunction, and interjection.

Q11. Differentiate open class and closed class words.

Open classes (noun, verb, adjective, adverb) readily admit new words and are large; closed classes (pronouns, prepositions, conjunctions, determiners) are fixed, small sets that rarely gain new members.

Q12. What is Part-of-Speech (POS) tagging?

POS tagging is the process of assigning a grammatical category (tag) to each word in a sentence based on its definition and context, e.g., "book" tagged as NOUN or VERB depending on usage.

Q13. Explain rule-based POS tagging.

Rule-based tagging uses a dictionary to assign possible tags to each word, then applies a set of hand-written disambiguation rules (based on context) to eliminate incorrect tags and select the appropriate one.

Q14. Explain stochastic POS tagging.

Stochastic tagging assigns tags based on probability, typically choosing the tag sequence that maximizes $P(\text{tag sequence} \mid \text{word sequence})$ using statistical models such as Hidden Markov Models trained on tagged corpora.

Q15. Explain Transformation-Based (Brill) tagging.

Brill tagging starts by assigning the most likely tag to each word, then applies an ordered set of learned transformation rules that correct tags based on context, iteratively improving tagging accuracy.

Q16. List two issues/challenges in POS tagging.

Ambiguous words with multiple possible tags (e.g., "back" as noun/verb/adjective/adverb), unknown/out-of-vocabulary words, and tagging errors from context-dependent usage.

Q17. What is a Hidden Markov Model (HMM)? Name its key components.

An HMM is a statistical model with hidden states (tags) generating observable outputs (words), defined by states, an observation vocabulary, transition probabilities, emission probabilities, and initial state probabilities.

Q18. State the two key assumptions used by an HMM POS tagger.

- (1) Markov assumption: a tag's probability depends only on the previous tag (limited history).
- (2) Output independence assumption: a word's probability depends only on its own tag, not on surrounding words/tags.

Q19. What is the Viterbi algorithm used for?

The Viterbi algorithm is a dynamic programming algorithm used to find the most probable sequence of hidden states (tags) given an observation sequence (words) in an HMM.

Q20. What is Maximum Entropy (MaxEnt) modeling in POS tagging?

MaxEnt is a discriminative probabilistic model that assigns tags using weighted feature functions (e.g., word suffix, capitalization, previous tag) combined log-linearly, choosing the tag distribution with maximum entropy that satisfies the feature constraints.

Q21. Differentiate generative and discriminative models with reference to HMM and MaxEnt.

A generative model (HMM) models the joint probability $P(\text{words, tags})$ and uses Bayes' rule to find the best tag sequence; a discriminative model (MaxEnt) directly models $P(\text{tags} \mid \text{words})$ using arbitrary overlapping features.

Q22. What is data sparsity in the context of n-gram models?

Data sparsity refers to the problem where many possible n-grams do not appear (or appear rarely) in even a large training corpus, leading to unreliable probability estimates for unseen sequences.

Q23. What is a bigram language model? Give its probability formula.

A bigram model approximates the probability of a word given all previous words using only the immediately preceding word: $P(w_n \mid w_1 \dots w_{n-1}) \approx P(w_n \mid w_{n-1})$.

Q24. What is tag ambiguity? Give an example.

Tag ambiguity occurs when a word can take more than one POS tag depending on context; e.g., "run" can be a VERB ("I run daily") or a NOUN ("a home run").

Q25. What is the role of a tagged corpus (e.g., Penn Treebank) in POS tagging?

A tagged corpus provides gold-standard word–tag pairs used to estimate transition/emission probabilities (for HMM) or train feature weights (for MaxEnt), and to evaluate tagger accuracy.

Part B (16 Marks)

Q1. Explain unsmoothed n-gram models and describe how they are evaluated.

Hint: Define n-gram, MLE formula, chain rule/Markov approximation; explain perplexity and cross-entropy as evaluation metrics with a worked numeric example; mention train/test split practice.

Q2. Explain various smoothing techniques used in language modeling (Add-one, Good-Turing, Kneser-Ney).

Hint: Motivate zero-probability problem; explain Add-one with formula; explain Good-Turing re-estimation; briefly explain Kneser-Ney (absolute discounting + continuation probability); compare pros/cons.

Q3. Explain interpolation and backoff techniques for combining n-gram models with suitable examples.

Hint: Define both; give linear interpolation formula with lambda weights; explain Katz backoff with discounting; illustrate with a trigram/bigram/unigram combination example.

Q4. Explain rule-based, stochastic, and transformation-based POS tagging approaches with examples.

Hint: Describe each approach's mechanism (dictionary+rules; HMM probability; Brill's learned transformation rules); compare accuracy, need for training data, and interpretability; give a short example sentence tagged by each method conceptually.

Q5. Explain the Hidden Markov Model for POS tagging and describe the Viterbi algorithm with an example.

Hint: Define HMM components (states, transition/emission matrices); state Markov & output-independence assumptions; walk through Viterbi trellis construction for a short sentence (e.g., "the dog barks") and identify the best tag path.

Q6. Explain Maximum Entropy Markov Models (MEMM) / MaxEnt models for POS tagging.

Hint: Explain log-linear model, feature functions (word shape, prefix/suffix, previous tag), training via maximum likelihood/GIS, and how it overcomes HMM's independence limitations; note the label-bias problem.

Q7. Discuss the issues and challenges faced in Part-of-Speech tagging.

Hint: Ambiguity, unknown words, idioms/multi-word units, domain/genre variation, tagset granularity differences, error propagation to downstream tasks; discuss mitigation strategies.

Q8. Compare and contrast the different POS tagging approaches (rule-based, stochastic/HMM, transformation-based, MaxEnt) in terms of accuracy, training data requirement, and computational complexity.

Hint: Build a comparison covering data needs, interpretability, handling of unknown words, accuracy trends, and typical use cases; conclude with which approach suits which scenario.

UNIT III — SYNTACTIC ANALYSIS

Part A — Two Mark Questions with Answers

Q1. Define Context-Free Grammar (CFG).

A CFG is a 4-tuple (N, Σ, R, S) consisting of non-terminal symbols, terminal symbols, production rules of the form $A \rightarrow \beta$, and a start symbol, used to describe the syntactic structure of a language.

Q2. Write a simple CFG rule set for a noun phrase (NP).

$NP \rightarrow \text{Det } N \mid \text{Det Adj } N \mid \text{Pronoun}$; e.g., "the big dog" is derived as $NP \rightarrow \text{Det Adj } N \rightarrow \text{"the" "big" "dog"}$.

Q3. What is a parse tree?

A parse tree is a hierarchical tree representation showing how a grammar derives a given sentence, with the start symbol at the root, non-terminals as internal nodes, and terminals (words) as leaves.

Q4. Define a Treebank.

A treebank is a corpus of sentences annotated with syntactic (parse tree) structure, such as the Penn Treebank, used for training and evaluating statistical parsers.

Q5. What is meant by "Normal Form" of a grammar? Name one.

A normal form is a restricted, standardized format for grammar production rules that simplifies parsing algorithms. Chomsky Normal Form (CNF) restricts every rule to $A \rightarrow BC$ or $A \rightarrow a$.

Q6. State the rules of Chomsky Normal Form (CNF).

In CNF, every production is either of the form $A \rightarrow B C$ (two non-terminals) or $A \rightarrow a$ (a single terminal), with no ϵ -productions or unit productions (except possibly $S \rightarrow \epsilon$).

Q7. What is Dependency Grammar?

Dependency grammar represents sentence structure as binary asymmetric relations (dependencies) between a head word and its dependents, rather than phrase-structure constituents.

Q8. Differentiate phrase-structure (constituency) grammar and dependency grammar.

Constituency grammar groups words into nested phrases (NP, VP, etc.) with a tree of constituents; dependency grammar directly links words via head–dependent relations without intermediate phrasal nodes.

Q9. What is syntactic parsing?

Syntactic parsing is the process of analyzing a sentence to determine its grammatical structure with respect to a given grammar, producing a parse tree or dependency structure.

Q10. What is structural (syntactic) ambiguity? Give an example.

Structural ambiguity occurs when a sentence has more than one valid parse tree. Example: "I saw the man with a telescope" can attach the PP "with a telescope" to either "saw" or "the man".

Q11. What is meant by dynamic programming parsing? Name one such algorithm.

Dynamic programming parsing avoids recomputation by storing partial parse results (sub-trees) in a table for reuse; the CYK algorithm and Earley algorithm are examples.

Q12. What is shallow parsing (chunking)?

Shallow parsing, or chunking, identifies flat, non-recursive phrases (like NP or VP chunks) in a sentence without building a full, deeply nested parse tree.

Q13. Define Probabilistic Context-Free Grammar (PCFG).

A PCFG is a CFG where each production rule $A \rightarrow \beta$ is assigned a probability, such that the probabilities of all rules expanding a given non-terminal A sum to 1, allowing ranking of alternative parses.

Q14. How is the probability of a parse tree computed in a PCFG?

The probability of a parse tree is the product of the probabilities of all the individual production rules used in generating that tree.

Q15. What is the CYK (Cocke-Younger-Kasami) algorithm?

CYK is a bottom-up dynamic programming parsing algorithm that works on grammars in Chomsky Normal Form, filling a triangular table to determine whether/how a string can be derived from the grammar.

Q16. What is Probabilistic CYK?

Probabilistic CYK extends the CYK algorithm to PCFGs by storing, for each cell and non-terminal, the maximum probability sub-parse, enabling recovery of the most probable parse tree (Viterbi parsing).

Q17. What is a Probabilistic Lexicalized CFG?

A lexicalized PCFG augments each non-terminal/rule with head-word information, conditioning rule probabilities on lexical heads, which improves parsing accuracy by capturing word-specific syntactic preferences.

Q18. What is the independence assumption limitation of plain PCFGs?

Plain PCFGs assume rule probabilities are independent of the specific words involved and of the surrounding context, which fails to capture lexical preferences (e.g., attachment ambiguity depending on specific verbs/nouns).

Q19. Define a feature structure in syntactic analysis.

A feature structure is a set of attribute–value pairs (e.g., NUMBER: singular, PERSON: 3rd) used to represent grammatical properties of a constituent for constraint-based grammar formalisms.

Q20. What is unification of feature structures?

Unification is an operation that merges two feature structures into a single, more specific structure if their attribute values are compatible (do not conflict); it fails if any shared attribute has conflicting values.

Q21. Why are feature structures and unification useful in grammar?

They enforce agreement constraints (e.g., subject-verb number agreement) directly within the grammar formalism, ruling out ungrammatical combinations such as "He walk" (person/number mismatch).

Q22. What is meant by grammar rules for English (e.g., $S \rightarrow NP VP$)?

These are CFG production rules that formally capture the phrase-structure patterns of English sentences, e.g., $S \rightarrow NP VP$, $VP \rightarrow V NP$, $NP \rightarrow Det N$, describing how larger constituents are built from smaller ones.

Q23. Differentiate top-down and bottom-up parsing strategies.

Top-down parsing starts from the start symbol S and expands rules to match the input string; bottom-up parsing starts from the input words and combines them upward toward the start symbol S .

Q24. What is meant by "ambiguity" being a central problem in parsing?

A sentence may have exponentially many valid parse trees under a broad-coverage grammar; parsers must efficiently represent and rank these to select the most plausible structure, which is why probabilistic parsing is important.

Q25. What is the role of a probabilistic parser over a plain CFG parser?

While a plain CFG parser only checks grammaticality and enumerates all possible trees, a probabilistic parser (PCFG-based) ranks the alternative parse trees by likelihood and selects the most probable one, resolving ambiguity.

Part B (16 Marks)

Q1. Explain Context-Free Grammars and describe how they are used to write grammar rules for English.

Hint: Define CFG formally; give example rule set (S, NP, VP, PP rules); derive a parse tree for a sample sentence (e.g., "the dog chased the cat"); discuss generative capacity and limitations (e.g., cannot enforce agreement without extensions).

Q2. Explain Treebanks and Normal Forms of grammar with examples.

Hint: Define treebank, describe Penn Treebank bracket notation with an example; explain why normal forms (CNF) are needed for algorithms like CYK; show conversion of a simple grammar into CNF.

Q3. Explain Dependency Grammar and compare it with phrase-structure grammar.

Hint: Define dependency relations (head, dependent, labels like nsubj, dobj); draw a dependency graph for a sample sentence; compare representational differences and use cases (e.g., free-word-order languages).

Q4. Describe dynamic programming approaches to syntactic parsing, including the CYK algorithm, with a worked example.

Hint: Explain motivation (avoiding exponential re-parsing), convert a small grammar to CNF, fill the CYK table for a short sentence, and read off valid parses from the top cell.

Q5. Explain shallow parsing (chunking) and its applications, contrasting it with full syntactic parsing.

Hint: Define chunk grammar (regex over POS tags), show an example of NP/VP chunking on a sentence, discuss speed/accuracy trade-off vs. full parsing, and mention applications (information extraction, named entity boundaries).

Q6. Explain Probabilistic Context-Free Grammars (PCFG) and the Probabilistic CYK algorithm with an example.

Hint: Define PCFG and rule probability constraint; compute the probability of a sample parse tree as a product of rule probabilities; extend CYK to store max probabilities per cell (Viterbi parsing) and trace back the best parse.

Q7. Explain Probabilistic Lexicalized CFGs and discuss why lexicalization improves parsing accuracy.

Hint: Explain head-finding rules, lexical dependency between head words (e.g., verb-object preferences), give an attachment-ambiguity example resolved by lexical statistics, and note added computational cost.

Q8. Explain feature structures and the unification operation, and show how they resolve grammatical agreement.

Hint: Represent feature structures as attribute-value matrices (AVMs) for NUMBER, PERSON, AGREEMENT; demonstrate unification of two structures with a worked example; show unification failure when attributes conflict (e.g., "He walk").

UNIT IV — SEMANTICS AND PRAGMATICS

Part A — Two Mark Questions with Answers

Q1. What are the requirements for a meaning representation in NLP?

A meaning representation must support verifiability, unambiguous representation, canonical form (same meaning $\rightarrow$ same representation), support inference/reasoning, and expressiveness to capture variables and relations.

Q2. Define First-Order Logic (FOL) as used in semantic representation.

FOL is a formal logical system using constants, variables, predicates, functions, quantifiers ($\forall, \exists$), and connectives ($\wedge, \vee, \neg, \rightarrow$) to represent the meaning of sentences unambiguously.

Q3. Give an FOL representation for "Every student reads a book."

$\forall x (\text{Student}(x) \rightarrow \exists y (\text{Book}(y) \wedge \text{Reads}(x, y)))$.

Q4. What is a Description Logic (DL)?

Description Logic is a family of knowledge-representation formalisms, a decidable subset of FOL, used to describe concepts (classes), roles (relations), and individuals, forming the basis of ontologies (e.g., OWL).

Q5. What is syntax-driven semantic analysis?

Syntax-driven semantic analysis builds the meaning representation of a sentence compositionally, in parallel with its syntactic parse tree, by combining the semantic attachments of each grammar rule.

Q6. What is a semantic attachment (semantic rule) in a grammar?

A semantic attachment is a rule associated with a CFG production that specifies how to compute the meaning representation of the parent constituent from the meaning representations of its children (often via lambda calculus).

Q7. What is lambda calculus used for in semantic analysis?

Lambda calculus provides a notation for representing functions with unbound variables (e.g., $\lambda x. \text{Student}(x)$), allowing systematic, compositional combination of partial meanings during parsing.

Q8. Define word sense.

A word sense is one of the discrete meanings that a word can have; e.g., "bank" has senses like "financial institution" and "river bank".

Q9. What are the relations between word senses? Name any two.

Synonymy (different words, same meaning), antonymy (opposite meaning), hyponymy/hypernymy (is-a relation), and meronymy/holonymy (part-whole relation).

Q10. Define homonymy and polysemy. How do they differ?

Homonymy is when unrelated meanings share the same word form (e.g., "bank" - money vs. river); polysemy is when related meanings share a word form (e.g., "paper" - material vs. newspaper vs. academic article).

Q11. What are thematic roles? Give two examples.

Thematic roles describe the semantic relationship between a verb and its arguments, e.g., Agent (doer of action), Theme/Patient (entity affected), Instrument, Location, Source, Goal.

Q12. What are selectional restrictions?

Selectional restrictions are semantic constraints a verb places on the type of its arguments; e.g., the verb "eat" requires its subject to be animate and its object to be edible ("The cow eats grass" vs. "*The idea eats grass").

Q13. Define Word Sense Disambiguation (WSD).

WSD is the task of automatically determining which sense of an ambiguous word is being used in a given context.

Q14. Explain the supervised approach to WSD.

Supervised WSD trains a classifier (e.g., Naive Bayes, SVM) on a sense-tagged corpus using contextual features (surrounding words, POS, collocations) to predict the correct sense of a target word in new text.

Q15. Explain the dictionary/thesaurus-based approach to WSD (Lesk algorithm).

The Lesk algorithm disambiguates a word by comparing the dictionary definition (gloss) of each candidate sense with the words in the surrounding context, choosing the sense whose gloss has the greatest word overlap with the context.

Q16. Explain the bootstrapping approach to WSD.

Bootstrapping (e.g., Yarowsky's algorithm) starts with a small set of labeled seed examples for each sense, trains a classifier, uses it to label more instances confidently, and iteratively expands the training set (semi-supervised learning).

Q17. What is word similarity? Name one thesaurus-based measure.

Word similarity quantifies how close in meaning two words are. A thesaurus-based measure, such as path length in WordNet's hypernym hierarchy or Wu-Palmer similarity, computes similarity based on the taxonomy structure.

Q18. What is distributional similarity? State the distributional hypothesis.

Distributional similarity measures word similarity based on the contexts in which words occur, based on the distributional hypothesis: "words that occur in similar contexts tend to have similar meanings."

Q19. Name two vector-space measures used to compute distributional word similarity.

Cosine similarity and Jaccard similarity (also Dice coefficient), computed over co-occurrence/context feature vectors.

Q20. What is pragmatics? How does it differ from semantics?

Semantics studies literal, context-independent meaning of words/sentences; pragmatics studies how context, speaker intent, and world knowledge influence the interpretation of meaning in actual use.

Q21. What is a predicate in FOL-based semantic representation?

A predicate is a symbol representing a property of, or relation between, entities; e.g., Loves(John, Mary) states the relation "Loves" holds between John and Mary.

Q22. Differentiate quantifiers $\forall$ and $\exists$ with examples.

$\forall$ (universal quantifier) means "for all", e.g., $\forall x \text{ Student}(x) \rightarrow \text{Smart}(x)$ ("every student is smart"); $\exists$ (existential quantifier) means "there exists", e.g., $\exists x \text{ Student}(x) \wedge \text{Smart}(x)$ ("some student is smart").

Q23. What is the "one sense per discourse" heuristic used in WSD?

It is the observation that a word tends to keep the same sense throughout a single document/discourse, which bootstrapping algorithms exploit to propagate sense labels across occurrences.

Q24. Give an example of selectional restriction violation.

"The rock ate the sandwich" violates the selectional restriction that the subject of "eat" must be animate, since "rock" is inanimate.

Q25. What is meant by "compositionality of meaning"?

The principle of compositionality states that the meaning of a complex expression (sentence) is determined by the meanings of its constituent parts and the rules used to combine them.

Part B - (16 Marks)**Q1. Explain the requirements for meaning representation and describe First-Order Logic as a representation scheme with examples.**

Hint: List requirements (verifiability, canonical form, inference, expressiveness, unambiguity); introduce FOL syntax (terms, predicates, quantifiers, connectives); translate 2–3 English sentences into FOL formulas.

Q2. Explain syntax-driven semantic analysis and describe how semantic attachments are used to build meaning representations.

Hint: Explain compositional semantics with CFG + semantic rule pairs; use lambda calculus notation; walk through building the FOL representation of a simple sentence bottom-up along its parse tree.

Q3. Explain Description Logics and their role in knowledge representation, comparing them with First-Order Logic.

Hint: Define concepts, roles, individuals, and TBox/ABox; give example DL statements; discuss decidability trade-off vs. full FOL expressiveness; mention application to ontologies (OWL, Semantic Web).

Q4. Explain word senses, relations between senses, and the concepts of thematic roles and selectional restrictions with examples.

Hint: Define sense, homonymy/polysemy, synonymy/antonymy/hyponymy; list thematic roles with example sentences; explain selectional restriction with a violation example; discuss how these aid semantic interpretation.

Q5. Explain the supervised, dictionary/thesaurus-based, and bootstrapping approaches to Word Sense Disambiguation, comparing their strengths and weaknesses.

Hint: Describe each method's algorithm (feature-based classifier; Lesk gloss overlap; Yarowsky-style bootstrapping); compare data requirements, accuracy, and scalability; worked mini-example for Lesk's algorithm.

Q6. Explain Word Similarity computation using thesaurus-based and distributional methods.

Hint: Explain path-based/Wu-Palmer thesaurus measures over WordNet hierarchy with a worked example; explain distributional hypothesis, construct a co-occurrence vector, and compute cosine similarity between two example words.

Q7. Discuss pragmatics and its relationship to semantics, illustrating with examples where context changes interpretation.

Hint: Define pragmatics; contrast literal vs. intended meaning; discuss deixis, implicature, speech acts briefly; give example sentences whose meaning depends on context/speaker intent.

Q8. Explain the process of Word Sense Disambiguation with a complete worked example using the Lesk algorithm.

Hint: Pick an ambiguous word (e.g., "bank"), list its WordNet-style glosses, present a context sentence, compute word overlap for each sense's gloss vs. context, and select the sense with maximum overlap; discuss limitations (gloss sparsity) and refinements (Extended Lesk).

UNIT V — DISCOURSE ANALYSIS AND LEXICAL RESOURCES

Part A — Two Mark Questions with Answers

Q1. What is discourse in NLP?

Discourse refers to a coherent sequence of sentences or utterances (e.g., a paragraph, conversation, or document) that together convey meaning beyond what individual sentences express in isolation.

Q2. Define discourse segmentation.

Discourse segmentation is the task of dividing a text or conversation into topically or structurally coherent segments (e.g., separating a news article into its constituent sub-topics).

Q3. What is discourse coherence?

Coherence is the property by which the sentences/utterances in a discourse are meaningfully connected to one another, forming a unified, interpretable whole rather than a random collection of sentences.

Q4. List two types of coherence relations between sentences.

Result, explanation, elaboration, contrast, and parallel are examples of coherence relations (as in Rhetorical Structure Theory) that link adjacent discourse segments.

Q5. What is a reference phenomenon in discourse?

Reference phenomena occur when an expression (e.g., a pronoun or noun phrase) is used to refer to an entity already introduced or inferable in the discourse, requiring resolution to the correct referent.

Q6. Define anaphora. Give an example.

Anaphora is the use of an expression (typically a pronoun) that refers back to a previously mentioned entity (its antecedent). Example: "Ravi left his book. He forgot it at home." — "He" and "it" are anaphors.

Q7. What is anaphora resolution?

Anaphora resolution is the NLP task of automatically identifying the antecedent (referent) that an anaphoric expression, such as a pronoun, refers to within a discourse.

Q8. Explain the basic idea of the Hobbs algorithm for anaphora resolution.

The Hobbs algorithm performs a syntactic, breadth-first, left-to-right search over the parse trees of the current and preceding sentences, in a specified traversal order, to find the first noun phrase that agrees in number/gender with the pronoun.

Q9. Explain the Centering algorithm for anaphora/coreference resolution.

The Centering algorithm tracks a ranked list of "forward-looking centers" (salient entities) in each utterance and a "backward-looking center" (the current topic), preferring pronouns to refer to the highest-ranked, most salient entity to maintain discourse coherence.

Q10. Differentiate anaphora resolution and coreference resolution.

Anaphora resolution specifically links an anaphor (e.g., pronoun) to its antecedent; coreference resolution is the broader task of clustering all mentions (pronouns, names, definite noun phrases) in a text that refer to the same real-world entity.

Q11. What is cataphora?

Cataphora is the reverse of anaphora — a reference expression appears before its referent is explicitly mentioned; e.g., "Before he spoke, Ravi paused." ("he" refers forward to "Ravi").

Q12. What is the Porter Stemmer?

The Porter Stemmer is a widely used rule-based algorithm that removes common morphological/inflectional suffixes from English words in a series of ordered steps to produce a word's stem (e.g., "running" → "run").

Q13. What is a lemmatizer? How does it differ from a stemmer?

A lemmatizer reduces a word to its valid dictionary base form (lemma) using vocabulary and morphological/POS analysis, unlike a stemmer, which uses crude suffix-stripping rules that may not produce a real word.

Q14. What is the Penn Treebank?

The Penn Treebank is a large annotated corpus of English text (news articles) manually tagged with POS tags and syntactic parse tree structures, widely used as a standard training/evaluation resource in NLP.

Q15. What is Brill's Tagger?

Brill's Tagger is a transformation-based, rule-learning POS tagger that starts with a baseline tagging and iteratively applies learned correction rules to improve accuracy, combining the strengths of rule-based and corpus-based methods.

Q16. What is WordNet?

WordNet is a large lexical database of English that groups words into sets of synonyms called synsets, each expressing a distinct concept, and links synsets via semantic relations such as hypernymy and meronymy.

Q17. What is a synset in WordNet?

A synset is a set of one or more synonymous words/terms in WordNet that represent a single underlying lexical concept, e.g., {car, automobile, motorcar}.

Q18. What is PropBank?

PropBank (Proposition Bank) is a corpus annotated with predicate-argument structures, labeling verb arguments with semantic roles (Arg0, Arg1, etc.) to support semantic role labeling.

Q19. What is FrameNet?

FrameNet is a lexical resource based on frame semantics that documents semantic frames (structured scenarios), their frame elements (roles), and the words (lexical units) that evoke each frame.

Q20. What is an LSTM? Why is it used in NLP?

LSTM (Long Short-Term Memory) is a type of recurrent neural network architecture with gating mechanisms (input, forget, output gates) that can learn and retain long-range dependencies in sequential data, making it effective for language modeling, tagging, and translation tasks.

Q21. What problem of simple RNNs does LSTM solve?

LSTM addresses the vanishing/exploding gradient problem of simple RNNs, which prevents them from learning long-distance dependencies in sequences, by using a memory cell with gates to control information flow.

Q22. What is sentiment analysis?

Sentiment analysis is the NLP task of automatically identifying and classifying the opinion, emotion, or attitude (e.g., positive, negative, neutral) expressed in a piece of text.

Q23. What is Machine Translation (MT)?

Machine Translation is the task of automatically translating text or speech from one natural language (source) to another (target) using computational methods (rule-based, statistical, or neural).

Q24. What is a chatbot? Name one type.

A chatbot is a software application that conducts conversation with users via text or speech; types include rule-based (pattern-matching) chatbots and retrieval/generative AI-based (ML/DL-driven) chatbots.

Q25. List any two real-world applications of NLP besides sentiment analysis, MT, and chatbots.

Information extraction, text summarization, question answering, spam/email filtering, speech recognition, and named entity recognition.

Part B — (16 Marks)

Q1. Explain discourse segmentation and coherence with suitable examples.

Hint: Define segmentation (topic boundaries) and coherence; introduce cohesion devices (lexical repetition, connectives) vs. coherence relations (Rhetorical Structure Theory — elaboration, cause, contrast); illustrate with a short multi-sentence example.

Q2. Explain reference phenomena in discourse and describe the Hobbs algorithm for anaphora resolution with an example.

Hint: Define reference/anaphora/cataphora; describe Hobbs' tree-search traversal order step by step; apply it to a two-sentence example to resolve a pronoun to its antecedent.

Q3. Explain the Centering algorithm for coreference/anaphora resolution with an example discourse.

Hint: Define Cb (backward-looking center) and Cf (forward-looking centers) list; explain transition types (Continue, Retain, Shift); trace centers across 2–3 sentences of a sample discourse to show pronoun resolution.

Q4. Explain coreference resolution and differentiate it from anaphora resolution, discussing approaches used to solve it.

Hint: Define coreference chains/clusters; contrast with anaphora; briefly cover rule-based (Hobbs/Centering), supervised classification (mention-pair), and modern clustering approaches; give an example with multiple coreferent mentions.

Q5. Describe the key NLP lexical resources — Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank, and FrameNet — explaining the purpose of each.

Hint: Give a short definition, example, and typical use case for each resource; group them as (a) normalization tools, (b) annotated corpora, (c) lexical-semantic databases.

Q6. Explain WordNet's structure and semantic relations (synonymy, hypernymy, meronymy) with examples, and describe its applications.

Hint: Explain synsets, is-a hierarchy (hypernym/hyponym), part-whole (meronym/holonym), antonymy; draw a small hypernym chain example (e.g., poodle → dog → canine → animal); mention applications (WSD, similarity, IR).

Q7. Explain the role of LSTM networks in NLP and describe how Machine Learning/Deep Learning approaches are applied to NLP tasks.

Hint: Explain RNN limitations, LSTM gate mechanism (forget/input/output gates, cell state) at a conceptual level; describe applications — sequence labeling (POS/NER), language modeling, sentiment classification; briefly mention word embeddings as input representation.

Q8. Discuss real-world applications of NLP — Sentiment Analysis, Machine Translation, and Chatbots — explaining the underlying techniques used in each.

Hint: For sentiment analysis, discuss lexicon-based vs. ML-based classification; for MT, briefly contrast rule-based/statistical/neural MT; for chatbots, contrast rule-based pattern matching with retrieval/generative models; give one example scenario for each.